Aror University of Art, Architecture, Design & Sukkur

BS(Artificial Intelligence)

Fall-2026

Software Engineering

Course Title:	Software Engineering
Course Code:	CSC-206
Credit Hours:	(3+0)
Program:	BS (Artificial Intelligence) , BS (Cyber Security)
Semester:	3rd
Pre-Requisite:	None
Course Instructor:	Rashida Naz
Consultation Hours:	11:00am – 01:00pm
Electronic mail:	

Description:

This undergraduate course introduces students to the principles, processes, and practices involved in developing high-quality software systems. It covers the complete software development life cycle, including requirement engineering, software design, implementation, testing, maintenance, and project management. Students learn software development methodologies such as Agile and Scrum, along with software design such as UML, version control, software quality assurance, risk management, and professional software engineering practices.

The course emphasizes applying engineering principles to real-world software projects through practical activities and team-based development.

Aims and objectives:

- To introduce students to the fundamental principles and practices of software engineering.
- To develop an understanding of systematic and disciplined approaches to software development.
- To familiarize students with the Software Development Life Cycle (SDLC) and modern development methodologies
- Appreciate the wider engineering issues that form the background to developing complex and evolving software-intensive systems.
- Employ group working skills including general organization, planning and time management and inter-group negotiation.
- Capture, document and analyze requirements. Translate a requirements specification into an implementable design, following a structured and organized process.
- Formulate a testing strategy for a software system, employing techniques such as unit testing, test driven development and functional testing.
- Evaluate the quality of the requirements, analysis and design work done during the module
- Collaborate effectively in team-based software projects

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Assessment:

S. No	Assessment Activities	Percentage	Total Activities
1.	Mid Term Exam	30%	1
2.	Final Exam	40%	1
3.	Assignment / Quizzes/ Class Project / Viva of Assignments and Class Project	30%	Multiple

Course Learning Outcomes:

	Course Learning Outcomes (CLO)	Domain	GA	Assessment Tool
1	Understand the basics of software engineering and develop ability to select appropriate software process models to use in a particular situation	C2	GA2	Class Participation, Quizzes, Exams, Assignments
2	Develop ability to understand software requirement and software development models	C3	GA3	Quizzes, Exams, Assignments, Class Projects
3	Understand basics of software testing and evaluation along with recommended guidelines for effect conduct of both activities	C3	GA2	Class Participation, Quizzes, Exams, Assignments

Domains:

C=Cognitive, A=Affective, P=Psychomotor

Levels:

Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 6: Creating}
Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}
Psychomotor = {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}

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Course Outlines

Week	Lec #	Subtopics	Reference	Course % Covered
Week 01	Lec 01	1.1 Introduction to Software Engineering; Characteristics; Software Costs; Software Products	Chapter 01	2.08%
	Lec 02	1.2 Layered Technology; Process Framework; Framework Activities vs. Umbrella Activities	Chapter 01	4.17%
	Lec 03	1.3 The Essence of Practice; Software Myths: Developer, Customer & Manager	Chapter 01	6.25%
Week 02	Lec 04	2.1 Software Development Approaches	Chapter 02	8.33%
	Lec 05	Plan-Driven Approach	Chapter 02	10.42%
	Lec 06	Incremental Approach	Chapter 02	12.50%
Week 03	Lec 07	Iterative Approach	Chapter 02	14.58%
	Lec 08	Evolutionary Approach	Chapter 02	16.67%
	Lec 09	Component-Based Development	Chapter 02	18.75%
Week 04	Lec 10	2.2 Software Process Models; Waterfall Model	Chapter 02	20.83%
	Lec 11	Incremental Development	Chapter 02	22.92%
	Lec 12	Rapid Application Development (RAD)	Chapter 02	25.00%
Week 05	Lec 13	Spiral Model	Chapter 02	27.08%
	Lec 14	Agile Development Methodology	Chapter 02	29.17%
	Lec 15	Agile Manifesto; Agile Principles	Chapter 02	31.25%
Week 06	Lec 16	Scrum Model	Chapter 03	33.33%
	Lec 17	Extreme Programming (XP)	Chapter 02	35.42%
	Lec 18	3.4 Requirements Engineering: Introduction; Sources of Requirements; Importance; Root Causes of Project Failure & Success	Chapter 03	37.50%
Week 07	Lec 19	Goals of Software Development; Stakeholder Environment; Customer Types	Chapter 03	39.58%
	Lec 20	Requirements Engineering Process	Chapter 05	41.67%
	Lec 21	Requirements Management	Chapter 05	43.75%
Week 08	Lec 22	Types of Requirements: Functional, Non-Functional, Domain Requirements; Inverse Requirements; Design & Implementation	Chapter 05	45.83%

		Constraints		
	Lec 23	4.3 Unified Modeling Language (UML); Use Case Diagram	Chapter 06	47.92%
	Lec 24	Activity Diagram	Chapter 06	50.00%
Week 09	Lec 25	Class Diagram	Chapter 06	52.08%
	Lec 26	5.1 Software Testing; Verification vs. Validation; Who Tests Software?; Unit Testing	Chapter 17	54.17%
	Lec 27	Integration Testing; Top-Down vs. Bottom-Up Approach	Chapter 17	56.25%
Week 10	Lec 28	Regression Testing	Chapter 17	58.33%
	Lec 29	6.1 Software Project Management; Introduction to Project Management; Iron Triangle	Chapter 24	60.42%
	Lec 30	Software Management Distinctions	Chapter 24	62.50%
Week 11	Lec 31	Software Project Management Activities	Chapter 24	64.58%
	Lec 32	Management Activities	Chapter 24	66.67%
	Lec 33	Project Plan Supplements	Chapter 24	68.75%
Week 12	Lec 34	Project Plan Supplements / Planning Considerations	Chapter 24	70.83%
	Lec 35	The Planning Process	Chapter 24	72.92%
	Lec 36	Project Scheduling	Chapter 24	75.00%
Week 13	Lec 37	Scheduling Problems	Chapter 24	77.08%
	Lec 38	Critical Path Method (CPM)	Chapter 24	79.17%
	Lec 39	CPM Class Activity	Chapter 24	81.25%
Week 14	Lec 40	Estimation of Software Projects; Introduction; Software Measurement	Chapters 26 & 27	83.33%
	Lec 41	Basics of Software Estimation	Chapters 26 & 27	85.42%
	Lec 42	LOC-Based Estimation Techniques; Class Activity	Chapters 26 & 27	87.50%
Week 15	Lec 43	Function Point Techniques; Examples & Activities	Chapters 26 & 27	89.58%
	Lec 44	COCOMO (Constructive Cost Model)	Chapter 09	91.67%
	Lec 45	COCOMO class activity	—	93.75%
Week 16	Lec 46	Revision Class	—	95.83%
	Lec 47	Revision Class	Course Material	97.92%
	Lec 48	Final Revision / Course Review	—	100%

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Text Book

1. Software Engineering by IAN SOMMERVILLE, Latest Edition, Pearson Education Ltd.

Reference Material

1. Software Engineering, A Practitioner's Approach by ROGER S. PRESSMAN

CLOs	GA1 Academic Education	GA2 Knowledge for Solving Computing Problems	GA3 Problem Analysis	GA4 Design/Development of Solutions	GA5 Modern Tool Usage	GA6 Individual & Team Work	GA7 Communication	GA8 Computing Professionalism & Society	GA9 Ethics	GA10 Life-long Learning
CLO 1	0	1	0	0	0	0	0	0	0	0
CLO 2	0	0	1	0	0	0	0	0	0	0
CLO 3	0	1	0	0	0	0	0	0	0	0

Instructor Contact Details

For any course related quires registered students can contact via email Rashida.faculty@aror.edu.pk

Prepared by: _____ Verified by_____

HOD_____

Approvals

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